Program 7
Web application to Kubernetes using Minikube

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Steps
1. Start minikube

```
┌──(1rv24mc045_ryan㉿ ryan-kali)-[~/Documents/program7]
└─$ minikube start --driver=docker
😄 minikube v1.37.0 on Debian kali-rolling
✨  Using the docker driver based on user configuration
📌 Using Docker driver with root privileges
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚜 Pulling base image v0.0.48 ...
🔥 Creating docker container (CPUs=2, Memory=3900MB) ...
🐳 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔗 Configuring bridge CNI (Container Networking Interface) ...
🔎 Verifying Kubernetes components...
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🏄 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

2. Check status

```
┌──(1rv24mc045_ryan㉿ ryan-kali)-
└─$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

3. Create a .yaml file to define the deployment configurations

nginx-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 replicas: 2
 selector:
  matchLabels:
   app: nginx
 template:
  metadata:
   labels:
    app: nginx
  spec:
   containers:
    - name: nginx
     image: nginx:latest
     ports:
      - containerPort: 80
```

4. Create a deployment

```
┌──(1rv24mc045_ryan㉿ ryan-kali)-[~/Documents/program7]
└─$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

5. Check deployments

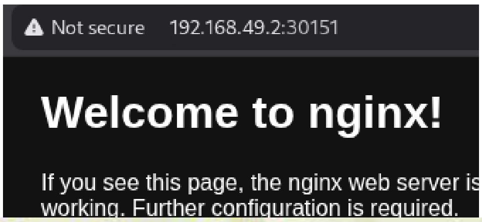
```
┌──(1rv24mc045_ryan㉿ ryan-kali)-[~/Documents/program7]
└─$ kubectl get pods
NAME                               READY  STATUS   RESTARTS  AGE
nginx-deployment-7d584c4448-2fchb  1/1    Running  0         14s
nginx-deployment-7d584c4448-r2pw2  1/1    Running  0         14s

┌──(1rv24mc045_ryan㉿ ryan-kali)-[~/Documents/program7]
└─$ kubectl get deployments
NAME              READY  UP-TO-DATE  AVAILABLE  AGE
nginx-deployment  2/2    2           2          18s
```

6. Expose the deployment as service

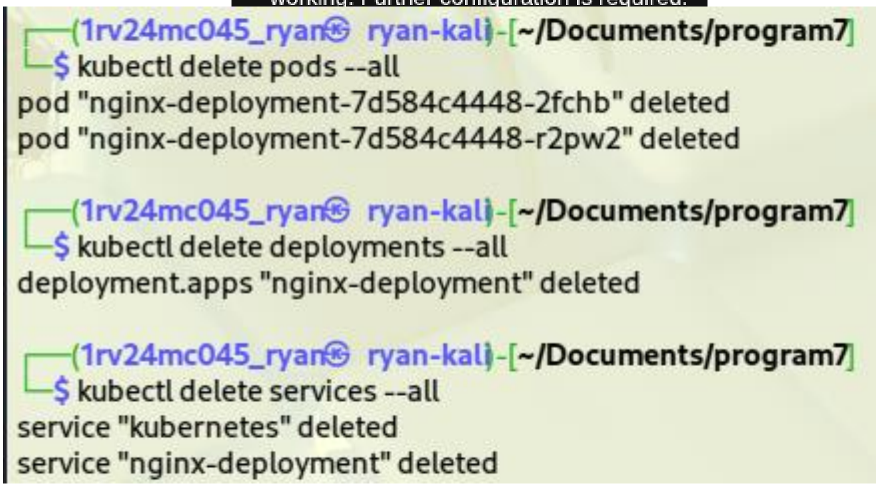
```
┌──(1rv24mc045_ryan㉿ ryan-kali)-[~/Documents/program7]
└─$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
```

7. Open the service in the browser to see the nginx deployment

```
┌──(1rv24mc045_ryan㉿ ryan-kali)-[~/Documents/program7]
└─$ minikube service nginx-deployment
|-----------|------------------|-------------|---------------------------|
| NAMESPACE |       NAME       | TARGET PORT |            URL            |
|-----------|------------------|-------------|---------------------------|
| default   | nginx-deployment | 80          | http://192.168.49.2:30151 |
|-----------|------------------|-------------|---------------------------|
🎉  Opening service default/nginx-deployment in default browser...
```



8. Delete all pods, services and deployments



```
┌──(1rv24mc045_ryan㉿ ryan-kali)-[~/Documents/program7]
└─$ kubectl delete pods --all
pod "nginx-deployment-7d584c4448-2fchb" deleted
pod "nginx-deployment-7d584c4448-r2pw2" deleted

┌──(1rv24mc045_ryan㉿ ryan-kali)-[~/Documents/program7]
└─$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted

┌──(1rv24mc045_ryan㉿ ryan-kali)-[~/Documents/program7]
└─$ kubectl delete services --all
service "kubernetes" deleted
service "nginx-deployment" deleted
```